sized_hdpe_500N

Hybrid Rocket Motor Design Report (HyProp)

Generated by HyProp

July 17, 2026

1 Mission Requirements

Required thrust: 500 N; total impulse: 4e+03 N s; design altitude: 1e+03 m; safety factor: 1.5.

2 Inputs and Assumptions

Oxidizer: N2O at 293 K (self-pressurized). Fuel: HDPE. Injector: 4×1.65 mm, C_d=0.6 (SPI). Efficiencies: η_{c^*}=0.95, nozzle extra η=0.9673, divergence half-angle 15.0°.

3 Grain and Chamber Geometry

Geometry: cylindrical; length 508 mm; outer diameter 43.1 mm; port parameters: $'\text{initial}_diameter'_m$: $0.025997243922216768.Tank : 2.4L, 1.66kgoxidizer.$

4 Transient Simulation

Burn time 17 s; total impulse 4.09e+03 N s; peak thrust 603 N; peak chamber pressure 38.1 bar; propellant used 2.05 kg; mean delivered I_{sp} 203 s. Events: liquid runout at 7.26 s; pressure decay at 17.04 s.

5 Representative Operating Points

The plain time average is misleading for blowdown motors: it understates thrust by 51% and chamber pressure by 48% versus the impulse-weighted point. All design modules therefore use the weighted points below. Time share needed to deliver impulse fractions: 50

Table 1: Representative operating points vs plain time average (key channels; full data in the XLSX export).

channel	peak_performance	impulse_weighted	heat_load	mass_flow	median_impulse	time_averag
F [N]	602.9	493.8	602.9	444.2	520.5	24
Pc [bar]	38.1	31.96	38.1	29.01	33.56	16.7
O/F [-]	6.443	5.359	6.443	4.915	5.535	3.00
Isp [s]	233.6	220.7	233.6	202.9	227.6	121.
c* [m/s]	1520	1488	1520	1455	1510	131
Tc [K]	3176	2963	3176	2807	3059	217
mdot_ox [kg/s]	0.2278	0.1885	0.2278	0.1708	0.1975	0.0966
mdot_f [kg/s]	0.03535	0.03468	0.03535	0.03277	0.03568	0.0239

Table 2: Burn phases: liquid / transition / gas.

phase	t0 [s]	t1 [s]	It [Ns]	It share	t share	F [N]	Pc [bar]	O/F
liquid	0	7.24	3764	0.9216	0.4254	519.8	33.49	5.556
transition	7.26	7.48	47.75	0.01169	0.01293	225	16.18	3.351
gas	7.5	17.02	260	0.06366	0.5593	27.45	3.978	1.052

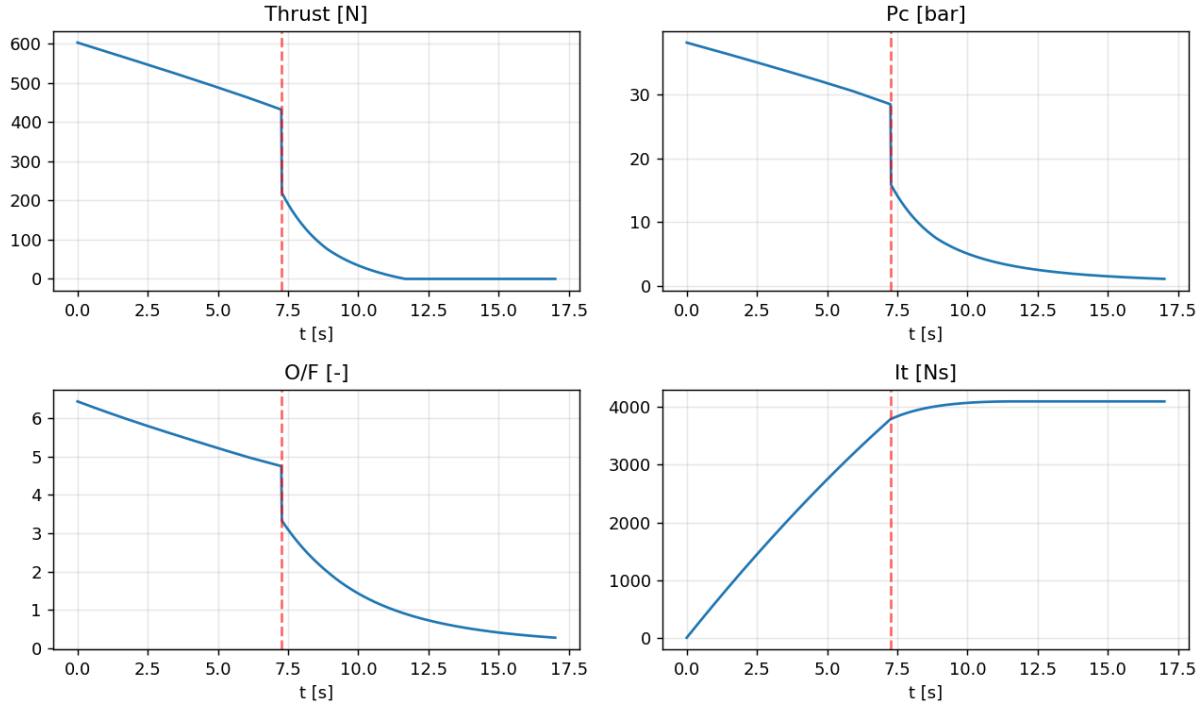
5.1 Recommendations

- Nozzle design and reported performance: **impulse-weighted** point (each instant weighted by its contribution to total impulse).
- Thermal/erosion analysis: **heat-load** point (maximum of the Bartz $\sim P_c^{0.8}$ proxy).
- Structural sizing and materials: **peak-performance** envelope (P_c=38.1 bar, T_c=3176 K).
- Injector and feed system: **mass-flow-weighted** point.

6 Nozzle and Performance

Throat diameter 11.7 mm; expansion ratio 5.02; impulse-weighted I_{sp} 221 s, c^* 1.49e+03 m/s, O/F 5.36.

7 Figures



8 Validation Note

Models validated against NASA-CEA-class thermochemistry (Cantera cross-check), HRAP transient benchmark (prusak_v2) and ESDU 91022/CoolProp N2O properties; see repository README. Empirical coefficients (a, n, C_d, η) require confirmation by fire tests before design freeze.